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Wireless Circuitry with Low-Noise Peak Detection

Abstract

An electronic device may include an antenna, a receiver, and a transmission line path that couples the antenna to the receiver. A peak detector may receive a radio-frequency signal from the transmission line path and may output a DC voltage indicative of a peak voltage level of the signal. The peak detector may include a pair of square law transistors having source-drain terminals coupled to an output of the peak detector. The peak detector may include a differential pair of transistors coupled between a power supply voltage and the output. The peak detector may include diode-connected transistors coupled between the source-drain terminals and gate terminals of the square law transistors. The diode-connected transistors may cause noise current to flow along feedback loop paths from the source-drain terminals onto the gate terminals of the square law transistors. This may remove noise from the output of the peak detector.

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Background/Summary

FIELD

[0001] This disclosure relates generally to electronic devices, including electronic devices with wireless communications circuitry.

BACKGROUND

[0002] Electronic devices are often provided with wireless communications capabilities. An electronic device with wireless communications capabilities has wireless circuitry with antennas that are used to transmit and receive radio-frequency signals.

[0003] Some radio-frequency signals are received by the electronic device at relatively low signal levels. If care is not taken, radio-frequency signals at relatively low signal levels can exhibit excessive noise, which can interfere with the satisfactory reception and measurement of the radio-frequency signals.

SUMMARY

[0004] An electronic device may include wireless circuitry. The wireless circuitry may include an antenna, a receiver, and a radio-frequency transmission line path that couples the antenna to the receiver. The wireless circuitry may include a peak detector coupled to the radio-frequency transmission line path. The peak detector may receive a radio-frequency signal from the radio-frequency transmission line path. The peak detector may output a DC voltage indicative of a peak voltage level of the radio-frequency signals (e.g., a DC voltage proportional to a square of the peak voltage level).

[0005] The peak detector may include a pair of square law transistors coupled in parallel between a power supply voltage and ground. The pair of square law transistors may have source-drain terminals coupled to an output of the peak detector. The peak detector may include a differential pair of transistors coupled between the power supply voltage and the output. The peak detector may include diode-connected transistors coupled between the source-drain terminals and gate terminals of the pair of square law transistors. The diode-connected transistors may cause noise current to flow in a single direction along feedback loop paths from the source-drain terminals onto the gate terminals of the pair of square law transistors. This may serve to cancel out noise from the DC voltage output by the peak detector.

[0006] An aspect of the disclosure provides a voltage peak detector having an input terminal and a differential output. The voltage peak detector can include a first transistor having a first gate terminal communicatively coupled to the input terminal, having a first source-drain terminal coupled to a first output terminal of the differential output, and having a second source-drain terminal communicatively coupled to a ground. The voltage peak detector can include a second transistor having a third source-drain terminal coupled to a power supply voltage and having a fourth source-drain terminal coupled to the first output terminal. The voltage peak detector can include a diode-connected transistor having a fifth source-drain terminal coupled to the first output terminal and having a sixth source-drain terminal coupled to the first gate terminal.

[0007] An aspect of the disclosure provides wireless circuitry. The wireless circuitry can include a radio-frequency transmission line path configured to convey a radio-frequency signal. The wireless circuitry can include an analog-to-digital converter (ADC). The wireless circuitry can include a peak detector having an output communicatively coupled to the ADC. The peak detector can include a differential pair of transistors coupled between a power supply voltage and the output. The peak detector can include a first transistor having a first gate terminal configured to receive the radio-frequency signal from the radio-frequency transmission line path and having a first source-drain terminal coupled to the output. The peak detector can include a second transistor having a second source-drain terminal coupled to the output and the first source-drain terminal, having a second gate terminal coupled to the second source-drain terminal, and having a third source-drain terminal coupled to the first gate terminal.

[0008] An aspect of the disclosure provides an electronic device. The electronic device can include

an antenna. The electronic device can include a receiver communicatively coupled to the antenna and configured to receive a radio-frequency signal using the antenna. The electronic device can include circuitry configured to identify a peak voltage level of the radio-frequency signal. The circuitry can include a pair of square law transistors coupled in parallel between a power supply voltage and a ground. The circuitry can include a differential pair of transistors coupled between the pair of square law transistors and the power supply voltage. The circuitry can include noise-cancelling devices coupled between source-drain terminals and gate terminals of the pair of square law transistors, the noise-cancelling devices being configured to pass current in a single direction from the source-drain terminals of the pair of square law transistors onto the gate terminals of the pair of square law transistors.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** is a diagram of an illustrative electronic device having wireless circuitry in accordance with some embodiments.

[0010] FIG. **2** is a diagram of illustrative wireless circuitry having a radio-frequency transmission line path in accordance with some embodiments.

[0011] FIG. **3** is a diagram of illustrative wireless circuitry having a peak detector coupled to a radio-frequency transmission line path in accordance with some embodiments.

[0012] FIG. **4** is a flow chart of illustrative operations involved in performing wireless communications using wireless circuitry having a peak detector coupled to a radio-frequency transmission line path in accordance with some embodiments.

[0013] FIG. **5** is a circuit diagram of an illustrative peak detector having noise-canceling transistors in accordance with some embodiments.

[0014] FIG. **6** is a plot of noise as a function of input power showing how an illustrative peak detector having noise-cancelling transistors may minimize noise in signal peak measurements in accordance with some embodiments.

DETAILED DESCRIPTION

[0015] Electronic device **10** of FIG. **1** may be a computing device such as a laptop computer, a desktop computer, a computer monitor containing an embedded computer, a tablet computer, a cellular telephone, a media player, or other handheld or portable electronic device, a smaller device such as a wristwatch device, a pendant device, a headphone or earpiece device, a device embedded in eyeglasses, goggles, a helmet, or other equipment worn on a user's head (e.g., a head-mounted display device), or other wearable or miniature device, a television, a computer display that does not contain an embedded computer, a gaming device, a navigation device, an embedded system such as a system in which electronic equipment with a display is mounted in a kiosk or automobile, a wireless internet-connected voice-controlled speaker, a home entertainment device, a remote control device, a gaming controller, a peripheral user input device, a wireless base station or access point, equipment that implements the functionality of two or more of these devices, or other electronic equipment.

[0016] As shown in the functional block diagram of FIG. **1**, device **10** may include components located on or within an electronic device housing such as housing **12**. Housing **12**, which may sometimes be referred to as a case, may be formed from plastic, glass, ceramics, fiber composites, metal (e.g., stainless steel, aluminum, metal alloys, etc.), other suitable materials, or a combination of these materials. In some embodiments, parts or all of housing **12** may be formed from dielectric or other low-conductivity material (e.g., glass, ceramic, plastic, sapphire, etc.). In other embodiments, housing **12** or at least some of the structures that make up housing **12** may be formed from metal elements.

[0017] Device **10** may include control circuitry **14**. Control circuitry **14** may include storage such as storage circuitry **16**. Storage circuitry **16** may include hard disk drive storage, nonvolatile memory (e.g., flash memory or other electrically-programmable-read-only memory configured to form a solid-state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Storage circuitry **16** may include storage that is integrated within device **10** and/or removable storage media.

[0018] Control circuitry **14** may include processing circuitry such as processing circuitry **18**. Processing circuitry **18** may be used to control the operation of device **10**. Processing circuitry **18** may include on one or more processors such as microprocessors, microcontrollers, digital signal processors, host processors, baseband processor integrated circuits, application specific integrated circuits, central processing units (CPUs), etc. Control circuitry **14** may be configured to perform operations in device **10** using hardware (e.g., dedicated hardware or circuitry), firmware, and/or software. Software code for performing operations in device **10** may be stored on storage circuitry **16** (e.g., storage circuitry **16** may include non-transitory (tangible) computer readable storage media that stores the software code). The software code may sometimes be referred to as program instructions, software, data, instructions, or code. Software code stored on storage circuitry **16** may be executed by processing circuitry **18**.

[0019] Control circuitry **14** may be used to run software on device **10** such as satellite navigation applications, internet browsing applications, voice-over-internet-protocol (VOIP) telephone call applications, email applications, media playback applications, operating system functions, etc. To support interactions with external equipment, control circuitry **14** may be used in implementing communications protocols. Communications protocols that may be implemented using control circuitry **14** include internet protocols, wireless local area network (WLAN) protocols (e.g., IEEE 802.11 protocols—sometimes referred to as Wi-Fi®), protocols for other short-range wireless communications links such as the Bluetooth® protocol or other wireless personal area network (WPAN) protocols, IEEE 802.11ad protocols (e.g., ultra-wideband protocols), cellular telephone protocols (e.g., 3G protocols, 4G (LTE) protocols, 5G protocols, etc.), antenna diversity protocols, satellite navigation system protocols (e.g., global positioning system (GPS) protocols, global navigation satellite system (GLONASS) protocols, etc.), antenna-based spatial ranging protocols (e.g., radio detection and ranging (RADAR) protocols or other desired range detection protocols for signals conveyed at millimeter and centimeter wave frequencies), or any other desired communications protocols. Each communications protocol may be associated with a corresponding radio access technology (RAT) that specifies the physical connection methodology used in implementing the protocol.

[0020] Device **10** may include input-output circuitry **20**. Input-output circuitry **20** may include input-output devices **22**. Input-output devices **22** may be used to allow data to be supplied to device **10** and to allow data to be provided from device **10** to external devices. Input-output devices **22** may include user interface devices, data port devices, and other input-output components. For example, input-output devices **22** may include touch sensors, displays (e.g., touch-sensitive and/or force-sensitive displays), light-emitting components such as displays without touch sensor capabilities, buttons (mechanical, capacitive, optical, etc.), scrolling wheels, touch pads, key pads, keyboards, microphones, cameras, buttons, speakers, status indicators, audio jacks and other audio port components, digital data port devices, motion sensors (accelerometers, gyroscopes, and/or compasses that detect motion), capacitance sensors, proximity sensors, magnetic sensors, force sensors (e.g., force sensors coupled to a display to detect pressure applied to the display), etc. In some configurations, keyboards, headphones, displays, pointing devices such as trackpads, mice, and joysticks, and other input-output devices may be coupled to device **10** using wired or wireless connections (e.g., some of input-output devices **22** may be peripherals that are coupled to a main processing unit or other portion of device **10** via a wired or wireless link).

[0021] Input-output circuitry **20** may include wireless circuitry **24** to support wireless

communications. Wireless circuitry **24** (sometimes referred to herein as wireless communications circuitry **24**) may include one or more antennas. Wireless circuitry **24** may also include baseband processor circuitry, transceiver circuitry, amplifier circuitry, filter circuitry, switching circuitry, radio-frequency transmission lines, and/or any other circuitry for transmitting and/or receiving radio-frequency signals using the antenna(s).

[0022] Wireless circuitry **24** may transmit and/or receive radio-frequency signals within a corresponding frequency band at radio frequencies (sometimes referred to herein as a communications band or simply as a “band”). The frequency bands handled by wireless circuitry **24** may include wireless local area network (WLAN) frequency bands (e.g., Wi-Fi® (IEEE 802.11) or other WLAN communications bands) such as a 2.4 GHz WLAN band (e.g., from 2400 to 2480 MHz), a 5 GHz WLAN band (e.g., from 5180 to 5825 MHz), a Wi-Fi® 6E band (e.g., from 5925-7125 MHz), and/or other Wi-Fi® bands (e.g., from 1875-5160 MHz), wireless personal area network (WPAN) frequency bands such as the 2.4 GHz Bluetooth® band or other WPAN communications bands, cellular telephone frequency bands (e.g., bands from about 600 MHz to about 5 GHz, 3G bands, 4G LTE bands, 5G New Radio Frequency Range 1 (FR1) bands below 10 GHz, 5G New Radio Frequency Range 2 (FR2) bands between 20 and 60 GHz, etc.), other centimeter or millimeter wave frequency bands between 10-300 GHz, near-field communications frequency bands (e.g., at 13.56 MHz), satellite navigation frequency bands (e.g., a Global Positioning System (GPS) L1 band (e.g., at 1575 MHz), L2 band (e.g., at 1228 MHz), L3 band (e.g., at 1381 MHz), L4 band (e.g., at 1380 MHz), and/or L5 band (e.g., at 1176 MHz), a Global Navigation Satellite System (GLONASS) band, a BeiDou Navigation Satellite System (BDS) band, etc.), ultra-wideband (UWB) frequency bands that operate under the IEEE 802.15.4 protocol and/or other ultra-wideband communications protocols, communications bands under the family of 3GPP wireless communications standards, communications bands under the IEEE 802.XX family of standards, satellite communications bands such as an L-band, S-band (e.g., from 2-4 GHz), C-band (e.g., from 4-8 GHz), X-band, Ku-band (e.g., from 12-18 GHz), Ka-band (e.g., from 26-40 GHz), etc. (e.g., for supporting bidirectional wireless communications via satellites orbiting Earth), industrial, scientific, and medical (ISM) bands such as an ISM band between around 900 MHz and 950 MHz or other ISM bands below or above 1 GHz, one or more unlicensed bands, one or more bands reserved for emergency and/or public services, and/or any other desired frequency bands of interest.

[0023] FIG. 2 is a diagram showing illustrative components within wireless circuitry **24**. As shown in FIG. 2, wireless circuitry **24** may include one or more processors such as processor(s) **26**, radio-frequency (RF) transceiver circuitry such as radio-frequency transceiver **28**, radio-frequency front end circuitry such as radio-frequency front end module (FEM) **40**, and antenna(s) **42**. Processor **26** may be a baseband processor, an application processor, a digital signal processor, a microcontroller, a microprocessor, a central processing unit (CPU), a programmable device, a combination of these circuits, and/or one or more processors within processing circuitry **18** (FIG. 1). Processor **26** may be configured to generate digital (transmit or baseband) signals. Processor **26** may be coupled to transceiver **28** over path **34** (sometimes referred to as a baseband path). Transceiver **28** may be coupled to antenna **42** via radio-frequency transmission line path **36**. Radio-frequency front end module **40** may be interposed on radio-frequency transmission line path **36** between transceiver **28** and antenna **42**.

[0024] Wireless circuitry **24** may include one or more antennas such as antenna **42**. Antenna **42** may be formed using any desired antenna structures. For example, antenna **42** may be an antenna with a resonating element that is formed from loop antenna structures, patch antenna structures, inverted-F antenna structures, slot antenna structures, planar inverted-F antenna structures, helical antenna structures, monopole antennas, dipoles, hybrids of these designs, etc. Two or more antennas **42** may be arranged into one or more phased antenna arrays (e.g., for conveying radio-frequency signals at millimeter wave frequencies). Parasitic elements may be included in antenna

42 to adjust antenna performance. Antenna **42** may be provided with a conductive cavity that backs the antenna resonating element of antenna **42** (e.g., antenna **42** may be a cavity-backed antenna such as a cavity-backed slot antenna).

[0025] In the example of FIG. 2, wireless circuitry **24** is illustrated as including only a single processor **26**, a single transceiver **28**, a single front end module **40**, and a single antenna **42** for the sake of clarity. In general, wireless circuitry **24** may include any desired number of processors **26**, any desired number of transceivers **28**, any desired number of front end modules **40**, and any desired number of antennas **42**. Each processor **26** may be coupled to one or more transceiver **28** over respective paths **34**. Each transceiver **28** may include a transmitter circuit configured to output uplink signals to antenna **42**, may include a receiver circuit configured to receive downlink signals from antenna **42**, and may be coupled to one or more antennas **42** over respective radio-frequency transmission line paths **36**. Each radio-frequency transmission line path **36** may have a respective front end module **40** disposed thereon. If desired, two or more front end modules **40** may be disposed on the same radio-frequency transmission line path **36**. If desired, one or more of the radio-frequency transmission line paths **36** in wireless circuitry **24** may be implemented without any front end module interposed thereon.

[0026] Front end module (FEM) **40** may include radio-frequency front end circuitry that operates on the radio-frequency signals conveyed (transmitted and/or received) over radio-frequency transmission line path **36**. Front end module may, for example, include front end module (FEM) components such as radio-frequency filter circuitry **44** (e.g., low pass filters, high pass filters, notch filters, band pass filters, multiplexing circuitry, duplexer circuitry, diplexer circuitry, triplexer circuitry, etc.), switching circuitry **46** (e.g., one or more radio-frequency switches), radio-frequency amplifier circuitry **48** (e.g., one or more power amplifiers and one or more low-noise amplifiers), impedance matching circuitry (e.g., circuitry that helps to match the impedance of antenna **42** to the impedance of radio-frequency transmission line **36**), antenna tuning circuitry (e.g., networks of capacitors, resistors, inductors, and/or switches that adjust the frequency response of antenna **42**), radio-frequency coupler circuitry, charge pump circuitry, power management circuitry, digital control and interface circuitry, and/or any other desired circuitry that operates on the radio-frequency signals transmitted and/or received by antenna **42**. Each of the front end module components may be mounted to a common (shared) substrate such as a rigid printed circuit board substrate or flexible printed circuit substrate. If desired, the various front end module components may also be integrated into a single integrated circuit chip.

[0027] Filter circuitry **44**, switching circuitry **46**, amplifier circuitry **48**, and other circuitry may be disposed on radio-frequency transmission line path **36**, may be incorporated into FEM **40**, and/or may be incorporated into antenna **42** (e.g., to support antenna tuning, to support operation in desired frequency bands, etc.). These components, sometimes referred to herein as antenna tuning components, may be adjusted (e.g., using control circuitry **14**) to adjust the frequency response and wireless performance of antenna **42** over time.

[0028] Radio-frequency transmission line path **36** may be coupled to an antenna feed on antenna **42**. The antenna feed may, for example, include a positive antenna feed terminal and a ground antenna feed terminal. Radio-frequency transmission line path **36** may have a positive transmission line signal path such that is coupled to the positive antenna feed terminal on antenna **42**. Radio-frequency transmission line path **36** may have a ground transmission line signal path that is coupled to the ground antenna feed terminal on antenna **42**. This example is illustrative and, in general, antennas **42** may be fed using any desired antenna feeding scheme. If desired, antenna **42** may have multiple antenna feeds that are coupled to one or more radio-frequency transmission line paths **36**.

[0029] Radio-frequency transmission line path **36** may include transmission lines that are used to route radio-frequency antenna signals within device **10** (FIG. 1). Transmission lines in device **10** may include coaxial cables, microstrip transmission lines, stripline transmission lines, edge-coupled microstrip transmission lines, edge-coupled stripline transmission lines, transmission lines

formed from combinations of transmission lines of these types, etc. Transmission lines in device **10** such as transmission lines in radio-frequency transmission line path **36** may be integrated into rigid and/or flexible printed circuit boards. In one suitable arrangement, radio-frequency transmission line paths such as radio-frequency transmission line path **36** may also include transmission line conductors integrated within multilayer laminated structures (e.g., layers of a conductive material such as copper and a dielectric material such as a resin that are laminated together without intervening adhesive). The multilayer laminated structures may, if desired, be folded or bent in multiple dimensions (e.g., two or three dimensions) and may maintain a bent or folded shape after bending (e.g., the multilayer laminated structures may be folded into a particular three-dimensional shape to route around other device components and may be rigid enough to hold its shape after folding without being held in place by stiffeners or other structures). All of the multiple layers of the laminated structures may be batch laminated together (e.g., in a single pressing process) without adhesive (e.g., as opposed to performing multiple pressing processes to laminate multiple layers together with adhesive).

[0030] Transceiver **28** may include wireless local area network transceiver circuitry that handles WLAN communications bands, wireless personal area network transceiver circuitry that handles the 2.4 GHz Bluetooth® band or other WPAN communications bands, cellular telephone transceiver circuitry that handles cellular telephone bands, near-field communications (NFC) transceiver circuitry that handles near-field communications bands, satellite navigation receiver circuitry that handles satellite navigation bands, ultra-wideband (UWB) transceiver circuitry that handles communications using the IEEE 802.15.4 protocol and/or other ultra-wideband communications protocols, and/or any other desired radio-frequency transceiver circuitry for covering any other desired communications bands of interest.

[0031] In performing wireless transmission, processor **26** may provide digital signals to transceiver **28** over path **34**. Transceiver **28** may further include circuitry for converting the baseband signals received from processor **26** into corresponding intermediate frequency or radio-frequency signals. For example, transceiver **28** may include mixer circuitry **50** for up-converting (or modulating) the baseband signals to intermediate frequencies or radio frequencies prior to transmission over antenna **42**. Transceiver **28** may also include digital-to-analog converter (DAC) and/or analog-to-digital converter (ADC) circuitry for converting signals between digital and analog domains. Transceiver **28** may include a transmitter component to transmit the radio-frequency signals over antenna **42** via radio-frequency transmission line path **36** and front end module **40**. Antenna **42** may transmit the radio-frequency signals to external wireless equipment by radiating the radio-frequency signals into free space.

[0032] In performing wireless reception, antenna **42** may receive radio-frequency signals from external wireless equipment. The received radio-frequency signals may be conveyed to transceiver **28** via radio-frequency transmission line path **36** and front end module **40**. Transceiver **28** may include circuitry for converting the received radio-frequency signals into corresponding intermediate frequency or baseband signals. For example, transceiver **28** may use mixer circuitry **50** for downconverting (or demodulating) the received radio-frequency signals to baseband frequencies prior to conveying the received signals to processor **26** over path **34**. Mixer circuitry **50** can include local oscillator circuitry such as local oscillator (LO) circuitry **52**. Local oscillator circuitry **52** can generate oscillator signals that mixer circuitry **50** uses to modulate transmitting signals from baseband frequencies to radio frequencies and/or to demodulate the received signals from radio frequencies to baseband frequencies.

[0033] The term “convey radio-frequency signals” as used herein means the transmission and/or reception of the radio-frequency signals (e.g., for performing unidirectional and/or bidirectional wireless communications with external devices). Similarly, the term “convey wireless data” as used herein means the transmission and/or reception of wireless data using radio-frequency signals (e.g., where the wireless data is modulated onto one or more carriers of the radio-frequency signals).

Antenna(s) **42** may transmit the radio-frequency signals by radiating the radio-frequency signals into free space (or to free space through intervening device structures such as a dielectric cover layer). Antenna(s) **42** may additionally or alternatively receive the radio-frequency signals from free space (or from free space through intervening device structures such as a dielectric cover layer). The transmission and reception of radio-frequency signals by antenna(s) **42** each involve the excitation or resonance of antenna currents on an antenna resonating element in the antenna(s) by the radio-frequency signals within the frequency band(s) of operation of the antenna(s).

[0034] In practice, it may be desirable to be able to measure the level of signal peaks in the radio-frequency signals conveyed over radio-frequency transmission line path **36** (e.g., between antenna(s) **42** and transceiver **28**). Wireless circuitry **24** may include a peak detector that measures the level of signal peaks in the radio-frequency signals conveyed over radio-frequency transmission line path **36**. FIG. **3** is a diagram showing one example of how wireless circuitry **24** may include a peak detector.

[0035] As shown in FIG. **3**, wireless circuitry **24** may include a signal peak detector such as peak detector **54** and a peak detector ADC such as ADC **56**. Peak detector **54** may have an input terminal coupled to radio-frequency transmission line path **36** over signal path **58**. Signal path **58** may be a single-ended signal path and the input of peak detector **54** may be a single-ended input that receives single-ended signals from radio-frequency transmission line path **36** over signal path **58**. Signal path **58** may be coupled to radio-frequency transmission line path **36** by a radio-frequency coupler, a signal splitter, a radio-frequency transformer, and/or other circuitry, as examples.

[0036] Peak detector **54** may have a differential output with a differential pair of output terminals coupled to a differential input of ADC **56** over differential signal path **60**. Differential signal path **60** may include a differential pair of coupled lines such as a first (positive) signal line **60P** and a second (negative) signal line **60N**. The differential pair of output terminals of peak detector **54** may include a first output terminal coupled to signal line **60P** and a second output terminal coupled to signal line **60N**. ADC **56** may have a digital output coupled to processor **26** (FIG. **3**), processing circuitry **18** (FIG. **1**), and/or other circuitry in device **10**.

[0037] Radio-frequency transmission line path **36** may convey radio-frequency signals between mixer circuitry **50** (FIG. **2**) and antenna(s) **42**. Signal path **58** and the input of peak detector **54** may be coupled to radio-frequency transmission line path **36** (e.g., at any desired location between antenna(s) **42** and mixer circuitry **50**). Signal path **58**, peak detector **54**, and ADC **56** may, as examples, be integrated into transceiver circuitry **50** or front end module **40** or may be separate from transceiver circuitry **50** and front end module **40** of FIG. **2**. While radio-frequency transmission line path **36** conveys radio-frequency signals, signal path **58** may tap or couple some of the radio-frequency signals (e.g., some of the power of the radio-frequency signals) off of radio-frequency transmission line path **36** and onto the input of peak detector **54**. Signal path **58** conveys the radio-frequency signals to the input of peak detector **54** as single-ended signals.

[0038] Peak detector **54** may perform measurements of the peak level (e.g., amplitude) of the radio-frequency signals received over signal path **58**. This may include measuring peak current level (e.g., peak current magnitude or current amplitude) and/or peak voltage level (e.g., peak voltage magnitude or voltage amplitude), as examples. Implementations in which peak detector **54** measures peak voltage levels (e.g., voltage amplitude) of the radio-frequency signals on radio-frequency transmission line path **36** are described herein as an example. Peak detector **54** is therefore sometimes also referred to herein as voltage peak detector **54**, peak voltage magnitude detector **54**, or voltage amplitude detector **54**.

[0039] Peak detector **54** may output a differential signal onto signal lines **60P** and **60N** that identifies the peak level of the radio-frequency signals on radio-frequency transmission line path **36**. Peak detector **54** may, for example, output a DC voltage across signal lines **60P** and **60N** having a magnitude that is based on the peak voltage level of the radio-frequency signals. ADC **56** may receive the DC voltage output by peak detector **54** over signal lines **60N** and **60P**. ADC **56** may

convert the DC voltage into a digital signal and may output the digital signal as voltage level information (VLI). Voltage level information VLI may, for example, be a digital code that identifies the magnitude of the DC voltage output by peak detector **54** and/or the peak voltage level of the radio-frequency signals on radio-frequency transmission line path **36** (as detected, estimated, measured, and/or output by peak detector **54**). Processing circuitry in device **10** may perform any desired operations based on voltage level information VLI (e.g., may adjust one or more tuning components of antenna(s) **42**, may adjust impedance matching circuitry, may switch an antenna into or out of use, may adjust a transmit power level of an antenna, may activate or deactivate a transceiver or antenna, may power one or more components of device **10** on or off, may adjust a power consumption mode of device **10**, may perform thermal mitigation operations, may identify and/or mitigate one or more jammer signals causing interference and/or noise on radio-frequency transmission line path **36**, may provide voltage level information VLI as an input to one or more software applications, etc.).

[0040] Some types of radio-frequency signals are received by the antenna(s) **42** in device **10** at relatively low signal levels. Such radio-frequency signals may include, for example, radio-frequency signals transmitted by one or more satellites in a satellite navigation band and/or a satellite communications band. These radio-frequency signals are received by device **10** at low signal levels because the corresponding transmitter is located relatively far from device **10** (e.g., on one or more satellites in space), subjecting the radio-frequency signals to substantial propagation/path loss in reaching device **10**. A transmitter may also have a limited output power level with which to transmit the radio-frequency signals. In some implementations, the transmitter may be a sensor node of a wireless sensor network. A sensor node in the wireless sensor network may be deployed in remote and/or harsh environments where communications links are subject to significant attenuation. In some implementations, the transmitter transmits the radio-frequency signals at relatively high frequencies that are subject to significant signal attenuation (e.g., millimeter wave frequencies, centimeter wave frequencies, sub-THz frequencies, etc.)

[0041] In these types of situations, it can be difficult for wireless circuitry **24** to differentiate a signal-of-interest in the radio-frequency signals (e.g., signal components that carry wireless data as transmitted by the corresponding transmitter) from noise. The noise may include thermal noise, electromagnetic noise, and/or other types of noise. At least some of the noise may, for example, be produced by one or more jammer signals present at device **10**. Such jammer signals may be produced by electronic components in device **10**, other antennas **42** in device **10**, transceiver circuitry in device **10**, nearby devices, thermal noise, and/or other sources of electromagnetic interference. If care is not taken, a low signal level of the signal-of-interest in the radio-frequency signals can cause noise to dominate the signal peak measurements performed by peak detector **54**. This can make it difficult for peak detector **54** to perform accurate signal peak measurements. If desired, peak detector **54** may include one or more noise-cancelling transistors that help to mitigate noise in the radio-frequency signals, allowing peak detector **54** to perform accurate peak level measurements even when the signal-of-interest in the radio-frequency signals is at a low signal level.

[0042] FIG. **4** is a flow chart of operations involved in performing peak level measurements using peak detector **54**. At operation **62**, antenna **42** may begin to receive radio-frequency signals. The radio-frequency signals may be conveyed, as one example, in a satellite band (e.g., a satellite navigation band or a satellite communications band). The radio-frequency signals may therefore contain a signal-of-interest at a relatively low signal level. Radio-frequency transmission line path **36** may pass the radio-frequency signals received by antenna **42** to a receiver in transceiver **28** (FIG. **2**).

[0043] At operation **64**, transceiver **28** may begin to receive the radio-frequency signals over radio-frequency transmission line path **36**. Transceiver **28** may begin to downconvert the radio-frequency signals to baseband signals (e.g., directly or via a first downconversion to an intermediate

frequency and then a second downconversion to baseband). Transceiver **28** may begin to attempt to decode/demodulate wireless data from the baseband signals. Wireless circuitry **24** may continue to perform operations **62** and **64** concurrent with the remaining operations of FIG. **4**.

[0044] At operation **66**, signal path **58** may pass a portion of the radio-frequency signals on radio-frequency transmission line path **36** to peak detector **54**. Peak detector **54** may measure the peak voltage level of the radio-frequency signals (e.g., may gather, generate, output, produce, estimate, and/or compute a peak voltage level value (measurement) that corresponds to or identifies the peak voltage level of the radio-frequency signals on radio-frequency transmission line path **36**). Peak detector **54** may, for example, generate a DC voltage across signal lines **60P** and **60N** having a magnitude that is proportional to or otherwise identifies the peak voltage level of the radio-frequency signals on radio-frequency transmission line path **36** (e.g., the magnitude may be proportional to a square of the peak voltage level). Peak detector **54** may include one or more noise-cancelling transistors that cancel out noise from the radio-frequency signals, helping to ensure the accuracy of the peak voltage level value output by peak detector **54**.

[0045] At operation **68**, ADC **56** may convert the voltage peak level measurement generated by peak detector **54** into a corresponding digital value (e.g., voltage level information VLI of FIG. **3**). ADC **56** may, for example, convert the DC voltage across signal lines **60P** and **60N** into a corresponding digital code that identifies the magnitude of the DC voltage. ADC **56** may pass the digital value to other circuitry for further processing.

[0046] At operation **70**, one or more processors in device **10** (e.g., processor **26** of FIG. **2** and/or processing circuitry **18** of FIG. **1**) may compute, calculate, estimate, or otherwise identify the peak voltage level of the radio-frequency signals based on the digital value (e.g., as proportional to a square of the DC voltage level identified by the digital value). The one or more processors in device **10** may perform any desired operations based on the peak voltage level of the radio-frequency signals (e.g., based on the digital value output by ADC **56** and thus the measurement performed by peak detector **54**). As one example, the one or more processors may identify a jammer signal in the radio-frequency signals and may take suitable action to mitigate the jammer signal based on the digital value output by ADC **56**. The one or more processors may, for example, identify that a jammer signal is present when/if the peak voltage level of the radio-frequency signals on radio-frequency transmission path exceeds a threshold peak voltage level. The one or more processors may attempt to mitigate the jammer signal by adjusting one or more tuning components of antenna(s) **42**, adjusting impedance matching circuitry, switching an antenna **42** into or out of use, adjusting a transmit power level of one or more of antennas **42**, activating or deactivating a transceiver or antenna in device **10**, powering one or more components of device **10** on or off, adjusting a power consumption mode of device **10**, performing thermal mitigation operations, and/or using any other methods. Once the jammer signal has been identified and mitigated, transceiver **28** may be able to more clearly identify the signal-of-interest in the received radio-frequency signals and to successfully decode wireless data from the signal-of-interest. As other examples, the one or more processors may detect the signal strength of the radio-frequency signals based on the digital value, may perform impedance measurements (e.g., S-parameter measurements) based on the digital value, may adjust one or more amplifiers, impedance matching circuitry, and/or antenna tuning circuitry on the radio-frequency transmission line path based on the digital value, may switch different antennas into or out of use based on the digital value, etc. These examples are illustrative and, in general, device **10** may perform any desired operations based on the measured peak voltage level.

[0047] FIG. **5** is a circuit diagram of peak detector **54**. As shown in FIG. **5**, peak detector **54** may include a differential signal path **80** that includes a first (positive) signal line **80P** and a second (negative) signal line **80N** coupled in parallel between power supply voltage VDD and a reference voltage such as ground **82**. Peak detector **54** may have an input terminal **72** (e.g., a single-ended input terminal) coupled to signal path **58** (FIG. **3**). Input terminal **72** may also be coupled to circuit

node **76** in peak detector **54**. If desired, peak detector **54** may include a capacitor **74** (e.g., a coupling capacitor) coupled in series between input terminal **72** and circuit node **54**.

[0048] Peak detector **54** may have a differential output **104** that includes a first (positive) output terminal **104P** coupled to signal line **80P** and that includes a second (negative) output terminal **104N** coupled to signal line **80N**. Output terminal **104P** is coupled to signal line **60P** and output terminal **104N** may be coupled to signal line **60N** of FIG. **3**.

[0049] Peak detector **54** may include a differential pair of transistors **90**, including a first transistor **90P** disposed on signal line **80P** and a second transistor **90N** disposed on signal line **80N**. The differential pair of transistors **90** may perform single-ended-to-differential signal conversion for peak detector **54**. Transistor **90P** has a first source-drain terminal communicatively coupled to power supply voltage **VDD** through a first segment of signal line **80P** and has a second source-drain terminal communicatively coupled to ground **82** through a second segment of signal line **80P**. The second source-drain terminal of transistor **90P** is also coupled to output terminal **104P**. The gate terminal of transistor **90P** is coupled to circuit node **88**.

[0050] Similarly, transistor **90N** has a first source-drain terminal communicatively coupled to power supply voltage **VDD** through a first segment of signal line **80N** and has a second source-drain terminal communicatively coupled to ground **82** through a second segment of signal line **80N**. The second source-drain terminal of transistor **90N** is also coupled to output terminal **104N**. The gate terminal of transistor **90N** is coupled to circuit node **88**. Peak detector **54** may include a resistor **92** coupled in series between circuit node **88** and output terminal **104P**. Peak detector **54** may also include a resistor **94** coupled in series between circuit node **88** and output terminal **104N**.

[0051] Peak detector **54** may include a pair of square law transistors such as transistors **78**, including a first transistor **78P** disposed on signal line **80P** and a second transistor **78N** disposed on signal line **80N**. Transistor **78P** has a first source-drain terminal coupled to output terminal **104P** and the second source-drain terminal of transistor **90P** (e.g., the first source-drain terminal of transistor **78P** may be communicatively coupled to power supply voltage **VDD** through transistor **90P**). Transistor **78P** has a second source-drain terminal coupled to ground **82** (e.g., through the second segment of signal line **80P**). The gate terminal of transistor **78P** may be coupled to circuit node **76** and may be communicatively coupled to input terminal **72** through circuit node **76** and capacitor **74**.

[0052] Transistor **78N** has a first source-drain terminal coupled to output terminal **104N** and the second source-drain terminal of transistor **90N** (e.g., the first source-drain terminal of transistor **78N** may be communicatively coupled to power supply voltage **VDD** through transistor **90N**). Transistor **78N** has a second source-drain terminal coupled to ground **82** (e.g., through the second segment of signal line **80N**). The gate terminal of transistor **78N** may be coupled to circuit node **98**. Transistors **78N** and **78P** are square law devices and may perform a squaring function on signals passing through the transistors along signal lines **80P** and **80N**.

[0053] Peak detector **54** may include a bias transistor **102** and a pair of bias resistors **100**, including bias resistor **100P** and bias resistor **100N**. Bias transistor **102** may have a first source-drain terminal coupled to its gate terminal. Bias transistor **102** may have a second source-drain terminal coupled to ground **82**. Bias resistor **100P** may be coupled in series between the first source-drain terminal of bias transistor **102** (or equivalently the gate terminal of bias transistor **102**) and circuit node **76** (or equivalently the gate terminal of transistor **78P**). Bias resistor **100N** may be coupled in series between the first source-drain terminal of bias transistor **102** (or equivalently the gate terminal of bias transistor **102**) and circuit node **98** (or equivalently the gate terminal of transistor **78N**). Bias transistor **102** may bias the gate terminals of transistors **78P** and **78N** during peak detection.

[0054] To help mitigate noise in the radio-frequency signals measured by peak detector **54**, peak detector **54** may include a pair of noise-cancelling transistors **96**, including a first noise-cancelling transistor **96P** and a second noise-cancelling transistor **96N**. Noise-cancelling transistor **96P** is a diode-connected transistor (e.g., having a source-drain terminal connected to its gate terminal) and

is coupled between output terminal **104P** and circuit node **76**. Noise-cancelling transistor **96P** is therefore sometimes also referred to herein as diode-connected transistor **96P**. Noise-cancelling transistor **96P** has a first source-drain terminal coupled to output terminal **104P** (or equivalently the first source-drain terminal of transistor **78P** and the second source-drain terminal of transistor **90P**). The first source-drain terminal of noise-cancelling transistor **96P**, output terminal **104P**, the second source-drain terminal of transistor **90P**, and the first source-drain terminal of transistor **78P** are coupled to the gate terminal of noise-cancelling transistor **96P**. Noise-cancelling transistor **96P** has a second source-drain terminal coupled to circuit node **76** (or equivalently the gate terminal of transistor **78P**). The current through diode-connected noise-cancelling transistor **96P** may flow in a single direction from the first source-drain terminal of noise-cancelling transistor **96P** to the second source-drain terminal of noise-cancelling transistor **96P**. Current may not flow in the opposite direction, from the second source-drain terminal to the first source-drain terminal of noise-cancelling transistor **96P**.

[0055] Noise-cancelling transistor **96N** is also a diode-connected transistor and is coupled between output terminal **104N** and circuit node **98**. Noise-cancelling transistor **96N** is therefore sometimes also referred to herein as diode-connected transistor **96N**. Noise-cancelling transistor **96N** has a first source-drain terminal coupled to output terminal **104N** (or equivalently the first source-drain terminal of transistor **78N** and the second source-drain terminal of transistor **90N**). The first source-drain terminal of noise-cancelling transistor **96N**, output terminal **104N**, the second source-drain terminal of transistor **90N**, and the first source-drain terminal of transistor **78N** are coupled to the gate terminal of noise-cancelling transistor **96N**. Noise-cancelling transistor **96N** has a second source-drain terminal coupled to circuit node **98** (or equivalently the gate terminal of transistor **78N**). The current through the diode-connected noise-cancelling transistor **96P** may flow in a single direction from the first source-drain terminal of noise-cancelling transistor **96N** to the second source-drain terminal of noise-cancelling transistor **96N**. Current does not flow in the opposite direction, from the second source-drain terminal to the first source-drain terminal of noise-cancelling transistor **96N**.

[0056] Implementations in which peak detector **54** includes noise-cancelling transistors **96P** and **96N** are described herein as an example. In general, noise-cancelling transistors **96P** and **96N** may be replaced with any desired types of noise-cancelling devices. As another example, noise-cancelling transistor **96P** may be replaced with a PN junction diode coupled between output terminal **104P** and circuit node **76** and/or noise-cancelling transistor **96N** may be replaced with a PN junction diode coupled between output terminal **104N** and circuit node **98**.

[0057] Transistors **90P**, **90N**, **78P**, **78N**, **96P**, **96N**, and **102** may be implemented using any desired transistor technologies. Transistors **90P**, **90N**, **78P**, **78N**, **96P**, **96N**, and/or **102** may include, for example, n-channel transistors such as n-type metal-oxide-semiconductor (NMOS) devices, p-channel transistors such as p-type metal-oxide-semiconductor (PMOS) devices, complementary metal-oxide-semiconductor (CMOS) devices, etc. Transistors **90P**, **90N**, **78P**, **78N**, **96P**, **96N**, and **102** have respective source, drain, and gate terminals. The terms “source” and “drain” are sometimes used interchangeably when referring to current-conducting terminals of a metal-oxide-semiconductor transistor. The source and drain terminals are therefore sometimes referred to as “source-drain” terminals (e.g., a transistor has a gate terminal, a first source-drain terminal, and a second source-drain terminal).

[0058] During signal peak measurement (e.g., while processing operation **66** of FIG. **4**), a portion of the radio-frequency signals on radio-frequency transmission line path **36** (FIG. **3**) is received as input signal RFIN at input terminal **72**. Input signal RFIN is a single-ended signal. Peak detector **54** may measure the peak voltage level of input signal RFIN and thus the radio-frequency signals on radio-frequency transmission line path **36**. Peak detector **54** may output a DC voltage VOUT across output terminals **104P** and **104N** that is indicative of the measured peak voltage level of input signal RFIN. Voltage VOUT is sometimes also referred to herein as output signal VOUT.

[0059] The magnitude of output signal VOUT (e.g., the DC voltage across output terminals **104P** and **104N**) is given by the difference between voltage VOUTP on output terminal **104P** and voltage VOUTN on output terminal **104N** (e.g., $|VOUT| = |VOUTP - VOUTN|$). If input signal RFIN has a peak voltage level VIN, the magnitude of output signal VOUT is proportional to the square of the absolute value of the peak voltage VIN of input signal RFIN (e.g., $|VOUTP - VOUTN| \sim |VIN|.sup.2$). Processing circuitry on device **10** may thereby identify the peak voltage VIN of the radio-frequency signals received over radio-frequency transmission line path **36** based on the magnitude of the output signal VOUT produced by peak detector **54** (e.g., after converting the magnitude of output signal VOUT into corresponding digital voltage level information VLI and accounting for the square law relation between the generated output signal VOUT and the magnitude VIN of input signal RFIN).

[0060] This relation between output signal VOUT and input signal RFIN is caused by the square law operation of transistors **78P** and **78N**. During peak measurement, input signal RFIN drives the gate terminal of transistor **78P** (e.g., while the gate terminals of transistors **78P** and **78N** are biased by bias transistor **102**). Corresponding current flows along signal line **80P** through transistor **78P** and along signal line **80N** through transistor **78N**, as shown by arrows **106**. The current flowing through transistors **78P** and **78N** may be decomposed into a signal current I.sub.S, associated with the signal-of-interest in input signal RFIN, and a noise current I.sub.N associated with noise in input signal RFIN. Transistors **78P** and **78N** apply a square law function to the voltage provided to output terminals **104P** and **104N**, producing voltage VOUTP on output terminal **104P** and voltage VOUTN on output terminal **104N** (e.g., outputting the output signal VOUT as a DC voltage across output terminals **104P** and **104N**). Transistors **90P** and **90N** perform single-ended-to-differential signal conversion on signal lines **80P** and **80N**, effectively converting from the single-ended signal domain of input signal RFIN to the differential signal domain of output signal VOUT.

[0061] Without noise-cancelling transistors **96P** and **96N** in peak detector **54**, output signal VOUT can be dominated by noise when the signal-of-interest in input signal RFIN is at a relatively low signal level. Noise-cancelling transistors **96P** and **96N** may form current feedback loops for transistors **78P** and **78N**, respectively, that serve to remove this noise from output signal VOUT.

[0062] For example, as shown by current loop **84** of FIG. 5, when the gate terminal of transistor **78P** is driven using input signal RFIN, the noise current I.sub.N flowing through the source-drain terminals of transistor **78P** passes from the first source-drain terminal of transistor **78P**, onto the first source-drain terminal of noise-cancelling transistor **96P**, through noise-cancelling transistor **96P** onto the second source-drain terminal of noise-cancelling transistor **96P**, and from the second source-drain terminal of noise-cancelling transistor **96P** onto circuit node **76** and the gate terminal of transistor **78P**. Because noise-cancelling transistor **96P** is diode-connected (e.g., because the same noise current I.sub.N is concurrently supplied to the gate terminal and the first source-drain terminal of noise-cancelling transistor **96P**), current does not flow in the reverse direction through noise-cancelling transistor **96P** and back onto the first source-drain terminal of transistor **78P**. The noise current I.sub.N that is driven onto the gate terminal of transistor **78P** by noise-cancelling transistor **96P** is inverted relative to the noise current I.sub.N flowing between the source-drain terminals of transistor **78P** (e.g., the noise current I.sub.N at the gate terminal of transistor **78P** and the noise current I.sub.N flowing through the source-drain terminals of transistor **78P** have opposite signs). This serves to substantially cancel out the noise current I.sub.N component of the voltage VOUTP produced at output terminal **104P** by transistor **78P**.

[0063] Similarly, as shown by current loop **86** of FIG. 5, the noise current I.sub.N flowing through the source-drain terminals of transistor **78N** passes from the first source-drain terminal of transistor **78N**, onto the first source-drain terminal of noise-cancelling transistor **96N**, through noise-cancelling transistor **96N** onto the second source-drain terminal of noise-cancelling transistor **96N**, and from the second source-drain terminal of noise-cancelling transistor **96N** onto circuit node **98** and the gate terminal of transistor **78N**. Because noise-cancelling transistor **96N** is diode-connected

(e.g., because the same noise current $I_{sub.N}$ is concurrently supplied to the gate terminal and the first source-drain terminal of noise-cancelling transistor **96N**), current does not flow in the reverse direction through noise-cancelling transistor **96P** and back onto the first source-drain terminal of transistor **78N**. The noise current $I_{sub.N}$ that is driven onto the gate terminal of transistor **78N** by noise-cancelling transistor **96N** is inverted relative to the noise current $I_{sub.N}$ flowing between the source-drain terminals of transistor **78N** (e.g., the noise current $I_{sub.N}$ at the gate terminal of transistor **78N** and the noise current $I_{sub.N}$ flowing through the source-drain terminals of transistor **78N** have opposite signs). This serves to substantially cancel out the noise current $I_{sub.N}$ component of the voltage V_{OUTN} produced at output terminal **104N** by transistor **78N**. In this way, noise-cancelling transistors **96P** and **96N** may cancel out noise (e.g., contributions from noise current $I_{sub.N}$) from input signal R_{FIN} while generating output signal V_{OUT} , preventing the noise from dominating output signal V_{OUT} and allowing peak detector **54** to generate highly accurate and noise-mitigated peak voltage level measurements of the radio-frequency signals on radio-frequency transmission line path **36**.

[0064] FIG. **6** is a plot showing how noise-cancelling transistors **96P** and **96N** may serve to minimize noise in output signals V_{OUT} . The vertical axis of FIG. **6** plots noise level (e.g., from $nV_{sup.2}$ to $\mu V_{sup.2}$). The horizontal axis of FIG. **6** plots the input power level $P_{sub.IN}$ of input signal R_{FIN} . Curve **110** plots the noise in output signal V_{OUT} in implementations where noise-cancelling transistors **96P** and **96N** are omitted from peak detector **54**. Curve **112** plots the noise in output signal V_{OUT} when peak detector **54** includes noise-cancelling transistors **96P** and **96N**. As shown by curves **110** and **112**, noise-cancelling transistors **96P** and **96N** may serve to reduce noise in output signal V_{OUT} across all input power levels $P_{sub.IN}$ of input signal R_{FIN} , by at least margin **114** (e.g., 10-30 dB). The example of FIG. **6** is illustrative and, in practice, curves **110** and **112** may have other shapes.

[0065] As used herein, the term “concurrent” means at least partially overlapping in time. In other words, first and second events are referred to herein as being “concurrent” with each other if at least some of the first event occurs at the same time as at least some of the second event (e.g., if at least some of the first event occurs during, while, or when at least some of the second event occurs). First and second events can be concurrent if the first and second events are simultaneous (e.g., if the entire duration of the first event overlaps the entire duration of the second event in time) but can also be concurrent if the first and second events are non-simultaneous (e.g., if the first event starts before or after the start of the second event, if the first event ends before or after the end of the second event, or if the first and second events are partially non-overlapping in time). As used herein, the term “while” is synonymous with “concurrent.” The term “when” also implies at least some concurrency (e.g., event A occurring “when” event B occurs means that at least some of event A is concurrent with at least some of event B).

[0066] The methods and operations described above in connection with FIGS. **1-6** may be performed by the components of device **10** using software, firmware, and/or hardware (e.g., dedicated circuitry or hardware). Software code for performing these operations may be stored on non-transitory computer readable storage media (e.g., tangible computer readable storage media) stored on one or more of the components of device **10** (e.g., storage circuitry **16** and/or wireless circuitry **24** of FIG. **1**). The software code may sometimes be referred to as software, data, instructions, program instructions, or code. The non-transitory computer readable storage media may include drives, non-volatile memory such as non-volatile random-access memory (NVRAM), removable flash drives or other removable media, other types of random-access memory, etc. Software stored on the non-transitory computer readable storage media may be executed by processing circuitry on one or more of the components of device **10** (e.g., processing circuitry in wireless circuitry **24**, processing circuitry **18** of FIG. **1**, etc.). The processing circuitry may include microprocessors, application processors, digital signal processors, central processing units (CPUs), application-specific integrated circuits with processing circuitry, or other processing circuitry.

[0067] Devices **10** may gather and/or use personally identifiable information. It is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.

[0068] The foregoing is illustrative and various modifications can be made to the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

Claims

1. A voltage peak detector having an input terminal and a differential output, the voltage peak detector comprising: a first transistor having a first gate terminal communicatively coupled to the input terminal, having a first source-drain terminal coupled to a first output terminal of the differential output, and having a second source-drain terminal communicatively coupled to a ground; a second transistor having a third source-drain terminal coupled to a power supply voltage and having a fourth source-drain terminal coupled to the first output terminal; and a diode-connected transistor having a fifth source-drain terminal coupled to the first output terminal and having a sixth source-drain terminal coupled to the first gate terminal.
2. The voltage peak detector of claim 1, wherein the diode-connected transistor has a second gate terminal that is connected to the fifth source-drain terminal.
3. The voltage peak detector of claim 1, wherein the diode-connected transistor is configured to pass a noise current from the fifth source-drain terminal onto the first gate terminal.
4. The voltage peak detector of claim 3, wherein the first transistor is configured to pass the noise current between the first source-drain terminal and the second source-drain terminal, the noise current as passed by the diode-connected transistor onto the first gate terminal being an inversion of the noise current as passed between the first source-drain terminal and the second source-drain terminal.
5. The voltage peak detector of claim 1, further comprising: a third transistor having a second gate terminal coupled to a first circuit node, having a seventh source-drain terminal coupled to a second output terminal of the differential output, and having an eighth source-drain terminal coupled to the ground.
6. The voltage peak detector of claim 5, further comprising: a fourth transistor having a ninth source-drain terminal coupled to the power supply voltage and having a tenth source-drain terminal coupled to the second output terminal, wherein the second transistor has a third gate terminal coupled to a second circuit node and the fourth transistor has a fourth gate terminal coupled to the second circuit node; a first resistor coupled in series between the second circuit node and the first output terminal; and a second resistor coupled in series between the second circuit node and the second output terminal.
7. The voltage peak detector of claim 6, further comprising: a fifth transistor having a fifth gate terminal, having an eleventh source-drain terminal coupled to the fifth gate terminal, and having a twelfth source-drain terminal, wherein the twelfth source-drain terminal is coupled to the ground; a third resistor coupled in series between the eleventh source-drain terminal and the first gate terminal; and a fourth resistor coupled in series between the eleventh source-drain terminal and the first circuit node.
8. The voltage peak detector of claim 7, further comprising: an additional diode-connected transistor having a thirteenth source-drain terminal coupled to the second output terminal, having a fourteenth source-drain terminal coupled to the first circuit node, and having a sixth gate terminal connected to the thirteenth source-drain terminal.

- 9.** The voltage peak detector of claim 5, further comprising: an additional diode-connected transistor having a ninth source-drain terminal coupled to the second output terminal and having a tenth source-drain terminal coupled to the first circuit node.
- 10.** The voltage peak detector of claim 9, further comprising: a fourth transistor having an eleventh source-drain terminal coupled to the power supply voltage and having a twelfth source-drain terminal coupled to the second output terminal, wherein the second transistor has a third gate terminal coupled to a fourth gate terminal of the fourth transistor.
- 11.** The voltage peak detector of claim 5, wherein the first transistor and the third transistor are square law devices.
- 12.** Wireless circuitry comprising: a radio-frequency transmission line path configured to convey a radio-frequency signal; an analog-to-digital converter (ADC); and a peak detector having an output communicatively coupled to the ADC, wherein the peak detector includes a differential pair of transistors coupled between a power supply voltage and the output, a first transistor having a first gate terminal configured to receive the radio-frequency signal from the radio-frequency transmission line path and having a first source-drain terminal coupled to the output, and a second transistor having a second source-drain terminal coupled to the output and the first source-drain terminal, having a second gate terminal coupled to the second source-drain terminal, and having a third source-drain terminal coupled to the first gate terminal.
- 13.** The wireless circuitry of claim 12, further comprising: a third transistor having a third gate terminal and having a fourth source-drain terminal, the fourth source-drain terminal being coupled to the output.
- 14.** The wireless circuitry of claim 13, further comprising: a fourth transistor having a fifth source-drain terminal coupled to the output and the fourth source-drain terminal, having a fourth gate terminal coupled to the fifth source-drain terminal, and having a sixth source-drain terminal coupled to the third gate terminal.
- 15.** The wireless circuitry of claim 14, further comprising: a fifth transistor having a seventh source-drain terminal communicatively coupled to the first gate terminal and the third gate terminal and having a fifth gate terminal coupled to the seventh source-drain terminal.
- 16.** The wireless circuitry of claim 12, wherein the peak detector is configured to generate a DC voltage at the output based on the radio-frequency signal, the ADC being configured to generate a digital value based on the DC voltage.
- 17.** The wireless circuitry of claim 16, wherein the radio-frequency signals are in a satellite navigation band and the wireless circuitry further comprises one or more processors configured to mitigate a jammer signal in response to the digital value exceeding a threshold value.
- 18.** An electronic device comprising: an antenna; a receiver communicatively coupled to the antenna and configured to receive a radio-frequency signal using the antenna; and circuitry configured to identify a peak voltage level of the radio-frequency signal, wherein the circuitry includes a pair of square law transistors coupled in parallel between a power supply voltage and a ground, a differential pair of transistors coupled between the pair of square law transistors and the power supply voltage, and noise-cancelling devices coupled between source-drain terminals and gate terminals of the pair of square law transistors, the noise-cancelling devices being configured to pass current in a single direction from the source-drain terminals of the pair of square law transistors onto the gate terminals of the pair of square law transistors.
- 19.** The electronic device of claim 18, wherein the noise-cancelling devices comprise diode-connected transistors.
- 20.** The electronic device of claim 18, wherein the noise-cancelling devices comprise diodes.
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